

This project focussed on developing a performance analytics framework using PlayerData athlete tracking data with the goal of helping coaches and trainers to better understand athlete workload and performance. While our team ended up producing a functional prototype that generated percentile-based performance norms, an important takeaway from this experience is how effectively we applied core course concepts such as design thinking, lean analytics and systems thinking as well as various structured analytical frameworks. Rather than just describing what we built as a team, this reflection evaluates how well these concepts were integrated into our process. In some areas like narrowing scope and translating outputs for the client we were able to apply these frameworks effectively. In other areas such as problem definition, data collection and overall planning we struggled a little bit. Understanding both sides is important because it highlights how stronger application of these concepts could have significantly improved both our process and the final product.

Design thinking emphasizes empathy, problem definition, ideation, prototyping and iteration. Our team demonstrated strength in the iteration aspect but was weaker in the early empathize and define stages. At the beginning of the project, our scope was way too broad. We were trying to analyze multiple sports, a bunch of different metrics and demographics all at once. Looking back, that clearly shows how we did not spend enough time really understanding what the client actually wanted at first. That changed after we started to meet with Buddy. His feedback helped us narrow everything down to just women's soccer and focus on a few key metrics like total distance, max speed and session load along with sprint events. This was probably the most important shift we made during the whole project. Once we made this adjustment, everything became more focused and easier to build around. From there, I think we did a much better job applying design thinking especially in terms of making our outputs simple and useful for the PlayerData team. We started thinking more about what a coach would actually want to see instead of just what we could analyze that would be the most interesting to us. This led us to use things like percentiles instead of just averages, which made the data a lot easier to understand. That being said, if we had fully applied design thinking from the beginning, we would have started with interviews or deeper stakeholder exploration rather than relying too much on our own assumptions. This likely would have saved time and led to a more focused approach earlier on in the process.

Lean analytics focuses on identifying key metrics that matter the most, testing them quickly and adjusting your approach based on feedback. This was one of the areas where our team showed some meaningful progress over time. Initially, we were trying to do too much and suffered from analysis overload by attempting to work with too many variables at once. However, once we finally narrowed our scope we aligned more closely with lean analytics principles by focusing on a small number of high-impact

metrics like total distance, max speed and session load which became the core drivers of our entire analysis. We were able to obtain some feedback along the way, particularly through our interactions with Buddy. His input helped us understand the metrics we were working with, identify the most important ones and decide how we should present them. An area where we might have fallen short was structured experimentation. While we adjusted based on feedback, we did not really test many different approaches or compare alternative ways of presenting these norms or defining benchmarks. Our process was more reactive than experimental. We could improve by explicitly defining hypotheses and testing these ideas both on our own and through feedback sessions. This would have made our application of lean analytics more intentional and rigorous.

Systems thinking is a holistic approach that involves understanding how different components of a project interact and influence one another. While the entire project structure we came up with was centered around this idea and our plan was executed fairly well, I still think there are areas where we could have improved. Our project involved multiple teams including Data Engineering, Data Science, Project Management and the team that I was a part of called the Business Analytics & Design Team. One of the main ideas within systems thinking is understanding interdependencies, which definitely showed up in our project. The work done by the Data Engineering team directly impacted what the Data Science team could analyze, which then affected what our Business Analytics & Design team could visualize and present. For example, when there were inconsistencies in the data structure between groups or missing variables like session type or position, it limited the types of insights we could generate later on. Even though we were all working toward the same goal, we did not always fully account for how decisions in one part of the project would impact others, especially earlier on. Another important piece of the puzzle is feedback loops, both reinforcing and balancing. We had some good reinforcing feedback loops, especially through our meetings with Buddy. His feedback helped us narrow our scope and improve our outputs to focus on what they were looking for at PlayerData, which then led to a better overall product in the end. We also dealt with some balancing feedback loops between teams. There were moments where earlier adjustments like changes in data structure or a shift in metric focus could have been communicated more clearly across groups to avoid rework or confusion. A stronger system of communication and iteration between teams would have made our process a little more efficient. While our project was built with systems thinking in mind, we could have applied it more intentionally by thinking a little more about dependencies earlier on and anticipating how changes in one area could impact the entire system.

Our project relied heavily on the PlayerData sample that was provided to us, but since the initial dataset was relatively small and limited in scope we also ended up generating some synthetic data to expand it. The idea behind this was to create a larger and more complete dataset that followed similar distributions and patterns as the original data, which would allow us to test our analytical framework at a bigger scale and create better visualizations. In a way, this synthetic data acted as a placeholder while we were waiting for Buddy to provide us with additional real data. It helped us continue building out our logic, testing calculations like percentiles and identifying potential issues early on. However, as the project progressed and we received updated datasets we ultimately decided not to rely on the synthetic data for our final analysis. This was mainly because synthetic data can be useful for testing but does not usually perfectly reflect variability that exists in the real world and could introduce misleading patterns if used for actual insights. From an analytical framework perspective, one of the strongest things the team did was develop percentile-based benchmarking which allowed us to evaluate athletes relative to their peers instead of just looking at raw numbers or averages. This made the analysis much more actionable and aligned with Buddy and PlayerData's needs. At the same time, our approach was still limited by factors like small and imbalanced sample sizes, missing contextual variables, and a narrow range of sports and age groups. These limitations definitely affected how reliable and generalizable our results were but I still think we maximized our efforts and did as much as we could with the dataset in the amount of time that we were given. Looking back, a stronger application of analytical frameworks would have included more focus on data quality and validation methods along with a clearer plan for how and when to use synthetic data appropriately within the overall analysis.

One of the strongest aspects of our project was turning benchmarking into something that is actually practical and usable rather than just another layer of analysis. Instead of only presenting averages or raw performance values, the data science team built a framework using percentile rankings that allows users to quickly understand how an athlete compares to others within their age group. This is important because these insights could be used to make real decisions. Coaches can use these benchmarks to adjust training intensity, monitor athlete workload and identify outliers or potential risks such as athletes consistently performing at very high or very low levels. We were also able to incorporate some of these findings into our visualizations by reshaping our focus and using solely box plots to compare the distributions of athletes based on certain performance metrics between age groups along with one scatterplot to demonstrate the relationship between weighted session load and maximum speed. As the project developed, we were able to take this a step further by integrating these benchmarks and visuals into an interactive dashboard which made the insights much easier to explore and interpret. Users could filter and view performance metrics dynamically

rather than relying on static tables or reports. The data engineering team also explored the use of a natural language chatbot that allows users to ask simple questions about performance and receive clear responses based on the data. While this demo that was presented at the end was more of a prototype, it showed how the benchmarking system and data visualizations could be made more accessible especially for non-technical users. Overall, this shift from static analysis to interactive and user-friendly tools was a major strength of the project even though there is still room to improve it in the future.

If we were to restart this project, I think the biggest difference would not just be what we actually come up with in the end but how we think about the problem from the start. Early on, we approached the project like a typical analytics assignment by taking the data and running analysis in order to find insights and create visualizations. This was the mindset that limited us and started us off on the wrong foot to begin with. We were focused more on what we could analyze that would be the most interesting to us rather than what actually mattered to the client. If we started over, the priority would be shifting from our original mindset that we were more familiar with as students to a mindset that is centered around PlayerData and what they are trying to accomplish. That would mean slowing down at the beginning and being more intentional about defining what success actually looks like before even touching the sample data or worrying about generating synthetic data while waiting for more data to be sent over. Another thing that would change is how we handle uncertainty. Throughout the project, we were often reacting to gaps in the data such as missing variables, imbalanced samples or unclear definitions rather than planning for them. If we restarted, we would treat those uncertainties as part of the system instead of obstacles. For example, instead of just working around missing context like session type or position, we could explicitly build our framework to account for incomplete data and clearly communicate those limitations from the beginning. This would make our analysis more honest and also more useful especially in a setting like this where the data is rarely perfect. We would also think more intentionally about how the final product is actually going to be used. A lot of our improvements came later when we started asking ourselves if a coach, trainer, athlete or anybody on the PlayerData team would actually use what we were providing them with and gain something meaningful from it. If we had that same perspective earlier on it would have changed how we built everything from the way we defined metrics to how we presented results. Instead of focusing first on calculations and then figuring out how to present them we would design the output first and then work backwards. This would ensure that everything we built directly supports decisions rather than just data analysis. Another shift would be taking more ownership of the full system in place instead of just our particular part in it. At times it felt like each team was focused on completing their own tasks rather than fully understanding how everything connected. If we restarted, there would be more emphasis on shared understanding across teams and helping each other out when one team has a lot more going on than

another. For example, the data engineering team did the majority of the work in the beginning but started to run out of things to do towards the end. It was kind of the opposite for us as members of the Business Analytics & Design team, with most of our work in the beginning being theoretical or just planning out what we would be doing later on. This does not necessarily mean we were not communicating effectively or planning correctly but I think that being more aware of how early decisions that were made could impact everything downstream would have been beneficial to us in the long run. The biggest lesson is that great data analysis is not just about getting the right answer but more about asking the right questions, understanding the system and making sure that the final product actually makes sense for the people who are using it.

More than anything, I think this project changed how I think about what it means to actually do data analytics for a company like PlayerData. Before this, I probably would have defined good work as having clean analysis, solid metrics and clear results that are presented using different types of visualizations. I see now that while all of these things are definitely important, they are not what makes this type of work valuable on their own. What really matters is whether the analysis makes sense for the situation that it is being used for based on who it's for, how it is interpreted and whether or not it is actually useful for the client by leading them to make better informed decisions. Going forward, I think I will try to approach these types of projects with a much different mindset. Instead of trying to prove that I can analyze data, I will be sure to stay focused on making sure that my work is useful from the start. This means being more comfortable slowing down early on even if it feels like I am not doing enough at first and making sure that I fully understand the context before building anything major. This project showed me that rushing into the analysis usually just leads to reworking things later anyway. It also made me realize that there is a difference between completing a task and actually being in charge of a project. There were times where it would have been easy to just focus on my role and move on, but the moments where the project improved the most were when we all stepped back and questioned the overall direction. In future work, I want to be more comfortable thinking beyond just my piece of the pie and considering how everything fits together. Another thing that I will take with me from this experience is just being okay with uncertainty. A lot of this project involved data related issues and constantly changing direction based on feedback. At times these felt like frustrating problems but they probably helped prepare us for what data analytics projects actually look like in the real world. Instead of trying to avoid that, I think the goal was to work through it in a way that still produced something useful in the end while learning everything that we did along the way to better prepare us for future work.